Year 12 Maths - Polynomials

Similar questions - Sketching cubics

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 In this question you must show detailed reasoning.

(a) Sketch the curve $y = (x+2)(x-1)^2$, showing the coordinates of all the points where it meets or touches the axes. **[4]**

(b) Find the exact area of the finite region enclosed between the curve and the x-axis. **[4]**

2 A cubic curve crosses the x-axis at $x = -1$, $x = 2$ and $x = 4$, and passes through the point $(0, 16)$.

(a) Find the equation of the curve in the form $y = k(x+1)(x-2)(x-4)$. **[3]**

(b) Express the equation in the form $y = ax^3 + bx^2 + cx + d$. **[2]**

(c) Find the set of values of x for which $y > 0$. **[3]**

Question 1

(a) Positive cubic. Crosses the x-axis at $(-2,0)$ and touches it at $(1,0)$ (a repeated root). Meets the y-axis where $y=(2)(1)=2$, at $(0,2)$

(b) $y=x^3-3x+2$. Area $=\displaystyle\int_{-2}^{1}\left(x^3-3x+2\right)\mathrm{d}x=\left[\frac{x^4}{4}-\frac{3x^2}{2}+2x\right]_{-2}^{1}=\frac{3}{4}-(-6)=\frac{27}{4}$

Question 2

(a) At $(0,16)$: $16=k(1)(-2)(-4)=8k$, so $k=2$ and $y=2(x+1)(x-2)(x-4)$

(b) $2(x+1)(x^2-6x+8)=2(x^3-5x^2+2x+8)=2x^3-10x^2+4x+16$

(c) Positive cubic with roots $-1,2,4$: $y>0$ for $-1<x<2$ or $x>4$